

CSC 177 - Assignment 2 Report

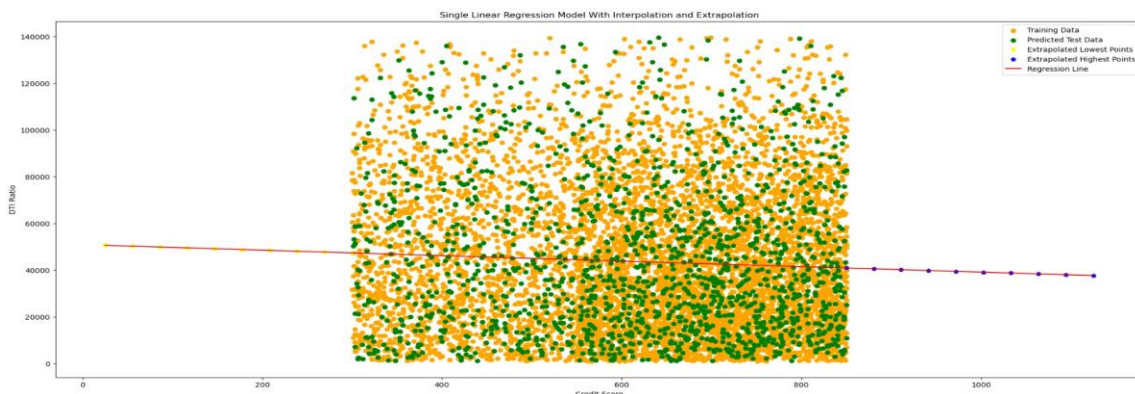
Pre-Part A - Preprocessing for the New Dataset

We chose to go ahead with a different dataset for this assignment. This dataset dealt with the approval odds of a loan of the applicant. With all the preprocessing done such as; counting and dropping any missing rows of data, removing the duplicate rows, removing outliers using the interquartile range method, converting the categorical column to numerical and plotting, we plotted the data. We also had to balance the dataset because as it was, about 80% of the data was leaning to the “reject” outcome which would not only heavily skew the data plot but also result in a poor model training. We did this by downsampling the “rejected” states to match the same number of dataset for the “approved” one. As a result, we got about 50-50 for both; “reject” and “approved” rows. We then split the dataset into 20% training and 80% testing model using the built-in library, “train_test_split”.

Part A - Regression Analysis (Single)

For the single linear regression model, we chose the “Credit_Score” column as the input for “x” values and “DTI_Ratio” for the observed values for “y” training the model. We then tested the x and y training model with their test models. These would be our interpolated data with the predicted y values from the model as no predicted y values would escape the minimum and maximum values of the observed data. Next, We tried to handle the extrapolation by first getting the minimum and maximum values of the training data. This was done so we could generate extrapolated points which would be below this minimum value as well as values which would be greater than the maximum. We attained this using *linspace*. We then combined the final input x as these two extrapolated points (as a list) as well as our original x inputs and then predicted the y values. We sorted the data before plotting to get a smooth regression line.

Result



The Root Mean Square Error = 31771.7144
The R-squared = 0.0042

Interpretation

For the interpolation data, we see a good scatter of the training and the test data's predicted values as they all fell in the range of the minimum and maximum "x" values with heavier clumps below the 50000 ratio. We can see an almost horizontal regression line which may signify that the linear relationship between the DTI_Score and the Credit_Score is a weak one. This may also be because of the dense scatter of the data where the line maybe capturing only little variance. Since there doesn't seem to be a strong linear relationship, we can interpret that as the credit score of one increases, it may not necessarily increase or decrease the DTI ratio of the applicant. We verified this by calculating the R-Squared value between these two variables and saw that it came to 0.0042 which is almost nil.

Using extrapolation, the same linear regression from the interpolation persists only with the extension of this line beyond the training and tested data values on both ends. This now helps us see the full regression line in that it is tipping in a downward slope. This extension will continue because we do not have any training data in those extreme regions so the model is holding a pure guess game that this trend will continue forever. In both the cases, the prediction denotes little to no correlation between the Credit Score and the DTI ratio.

Part A - Regression Analysis (Multiple)

For the multiple linear regression model, we chose the "Credit_Score" column as the target value. We then interpolated the data by choosing the remaining columns as the features and passed it for training and testing the model. For extrapolation, we did the same as above; however, to generate the values out of the range, we would require the mean values of the feature columns. This was needed for generating the values below and above the minimum and maximum training values, respectfully. We then tested the training model with their test models by supplying these values in bulk to get the predicted "y" values. We attained this again by using *linspace*. We then combined the final input "x" as these two extrapolated points (as a list) as well as our original "x" inputs and then predicted the "y" values. We sorted the data before plotting to get a smooth regression line.

Result



Interpretation

The interpolation was done the similar way to the single regression. We can see the predicted test data and the training data have overlaps which tells us that the model's predicted values were similar if not same to the observed data used for training it. We can see the downward slope of the regression line in the middle once again which suggests a slight negative relationship of income to the credit score. This makes sense logically that a higher income does not necessarily mean a better credit score but almost frequently, the lower the income, the lower the credit score of an individual. The calculated R-squared score also explains this correlation. It denotes that the model can explain 32% of the variance while other features are held constant and income alone is not the predominant feature in determining the credit score.

Extrapolation was done in a similar way too. Since our model was linear, the regression line was extended beyond the minimum and maximum data points. While holding other features constant using their mean values, the regression line allowed us to clearly see the trend in the regression line.

All in all, the overall results were pretty similar for both, the single as well as the multiple line regression with a noticeable change in the R-squared value between both.

Part B – Classification Tree Analysis (Smaller Dataset)

For this part, we used a smaller dataset different from the one in Part A. The process began by loading the dataset and cleaning the column names to remove any extra whitespace. The "Serial No." column was dropped since it was not required for analysis. Next, we defined our input features and the continuous target variable ("Chance of Admit") and then performed an 80/20 train-test split. This single split was used for both the regression and classification analyses. In addition, the features were normalized to support the multiple regression model.

Prashant Ram, Muhammad Hassan

Initial dataset preview:

	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	\
0	337	118	4	4.5	4.5	9.65	1	
1	324	107	4	4.0	4.5	8.87	1	
2	316	104	3	3.0	3.5	8.00	1	
3	322	110	3	3.5	2.5	8.67	1	
4	314	103	2	2.0	3.0	8.21	0	

Chance of Admit

0	0.92
1	0.76
2	0.72
3	0.80
4	0.65

Column names:

['GRE Score', 'TOEFL Score', 'University Rating', 'SOP', 'LOR', 'CGPA', 'Research', 'Chance of Admit']

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 500 entries, 0 to 499

Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	GRE Score	500 non-null	int64
1	TOEFL Score	500 non-null	int64
2	University Rating	500 non-null	int64
3	SOP	500 non-null	float64
4	LOR	500 non-null	float64
5	CGPA	500 non-null	float64
6	Research	500 non-null	int64
7	Chance of Admit	500 non-null	float64

dtypes: float64(4), int64(4)

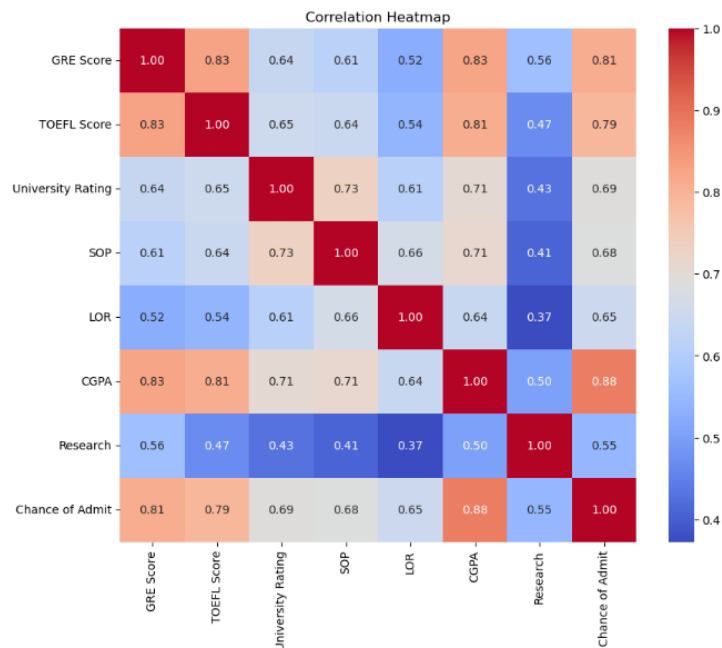
memory usage: 31.4 KB

None

Summary Statistics:

	GRE Score	TOEFL Score	University Rating	SOP	LOR	\
count	500.000000	500.000000	500.000000	500.000000	500.000000	
mean	316.472000	107.192000	3.114000	3.374000	3.484000	
std	11.295148	6.081868	1.143512	0.991004	0.92545	
min	290.000000	92.000000	1.000000	1.000000	1.000000	
25%	308.000000	103.000000	2.000000	2.500000	3.000000	
50%	317.000000	107.000000	3.000000	3.500000	3.500000	
75%	325.000000	112.000000	4.000000	4.000000	4.000000	
max	340.000000	120.000000	5.000000	5.000000	5.000000	

	CGPA	Research	Chance of Admit
count	500.000000	500.000000	500.000000
mean	8.576440	0.560000	0.72174
std	0.604813	0.496884	0.14114
min	6.800000	0.000000	0.34000
25%	8.127500	0.000000	0.63000
50%	8.560000	1.000000	0.72000
75%	9.040000	1.000000	0.82000
max	9.920000	1.000000	0.97000



2. Regression Analysis

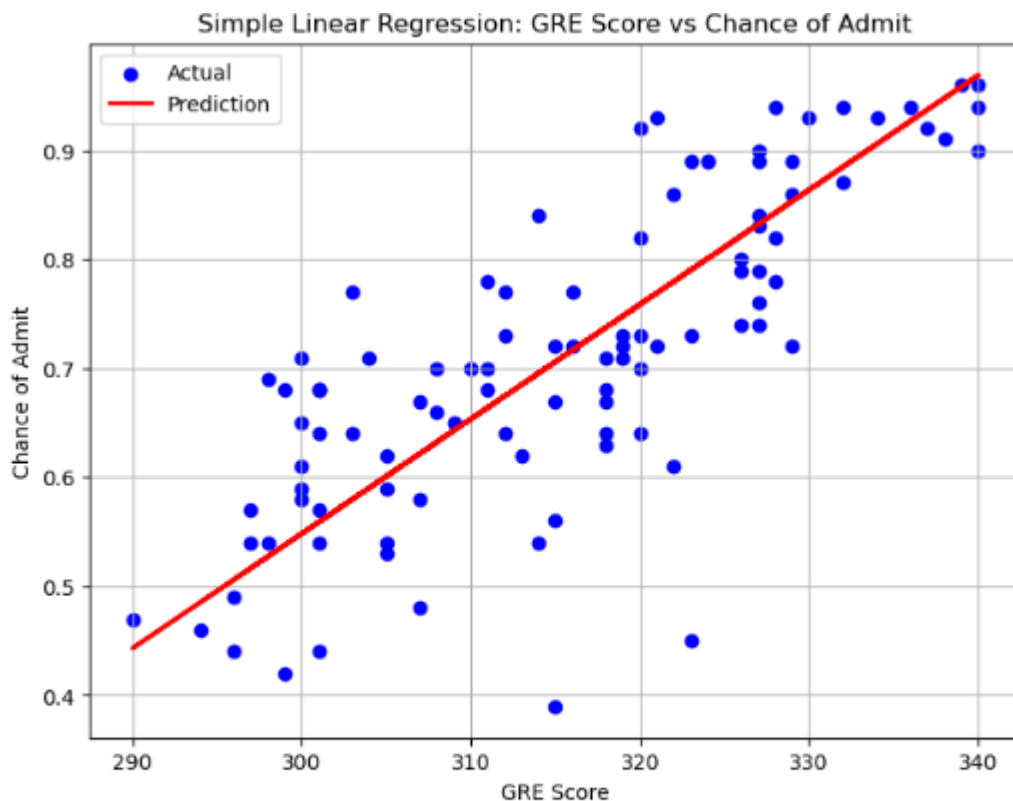
Although the primary focus of Part B is classification, we first performed regression analysis to provide context and compare continuous versus categorical outcomes.

2.1 Simple Linear Regression

A simple linear regression model was built using “GRE Score” as the sole predictor for “Chance of Admit.” The model’s predictions were visualized in a scatter plot where the regression line is overlaid on the actual data points. The performance of this model was evaluated using Mean Squared Error (MSE) and the R^2 score, both of which indicate that the predictive power of a single variable is limited.

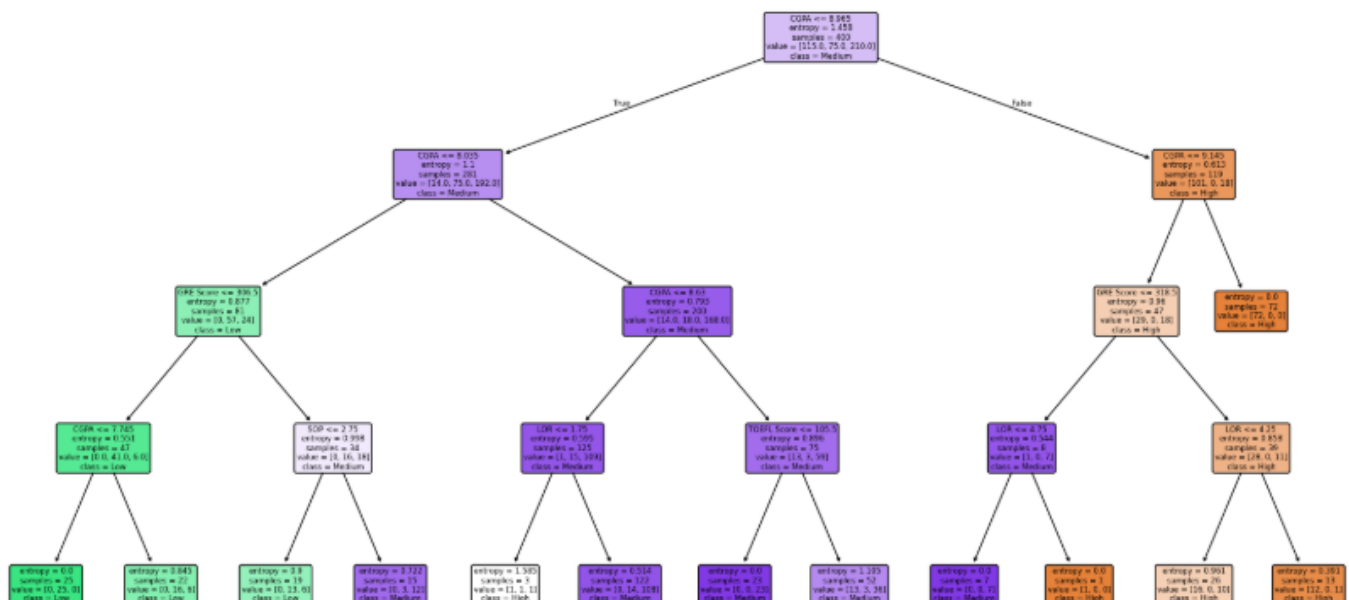
2.2 Multiple Linear Regression

Next, a multiple linear regression model was constructed using all available features (after scaling). Although this model offered slightly improved predictions, the performance metrics confirmed that the relationship remains complex and only partially explained by the input features.



Simple Linear Regression Metrics:
MSE: 0.008464325543124892
 R^2 Score: 0.5860965504584406

Multiple Linear Regression Metrics:
MSE: 0.0037046553987884114
 R^2 Score: 0.8188432567829628



The overall performance of the classifier was evaluated using accuracy on the test set. Additionally, decision rules were extracted from the tree, offering clear insights into the model's logic.

Extracted Decision Rules from the Classification Tree:

```

|--- CGPA <= 8.97
|   |--- CGPA <= 8.03
|   |   |--- GRE Score <= 306.50
|   |   |   |--- CGPA <= 7.75
|   |   |   |   |--- class: Low
|   |   |   |--- CGPA > 7.75
|   |   |   |   |--- class: Low
|   |   |--- GRE Score > 306.50
|   |   |   |--- SOP <= 2.75
|   |   |   |   |--- class: Low
|   |   |   |--- SOP > 2.75
|   |   |   |   |--- class: Medium
|   |--- CGPA > 8.03
|   |   |--- CGPA <= 8.63
|   |   |   |--- LOR <= 1.75
|   |   |   |   |--- class: High
|   |   |   |--- LOR > 1.75
|   |   |   |   |--- class: Medium
|   |   |--- CGPA > 8.63
|   |   |   |--- TOEFL Score <= 105.50
|   |   |   |   |--- class: Medium
|   |   |   |--- TOEFL Score > 105.50
|   |   |   |   |--- class: Medium
|--- CGPA > 8.97
|   |--- CGPA <= 9.14
|   |   |--- GRE Score <= 318.50
|   |   |   |--- LOR <= 4.75
|   |   |   |   |--- class: Medium
|   |   |   |--- LOR > 4.75
|   |   |   |   |--- class: High
|   |   |--- GRE Score > 318.50
|   |   |   |--- LOR <= 4.25
|   |   |   |   |--- class: High
|   |   |   |--- LOR > 4.25
|   |   |   |   |--- class: High
|   |--- CGPA > 9.14
|   |   |--- class: High

```

Classification Tree Homework Solution

1. ID3 Tree Construction:

a. Dataset:

ID	Color	Shape	Size	Class
1	Red	Square	Big	+
2	Blue	Square	Big	+
3	Red	Round	Small	-
4	Green	Square	Small	-
5	Red	Round	Big	+
6	Green	Round	Big	-

b. Initial entropy of dataset:

i. 3 positives and 3 negatives:

$$E(t) = - \left(\frac{3}{6} \log_2 \frac{3}{6} + \frac{3}{6} \log_2 \frac{3}{6} \right) = -2 \cdot (0.5 \cdot \log_2 0.5) = 1$$

1.

c. Information Gain:

i. Color:

1. Red = [+ , - , +] = Entropy = 0.918

2. Blue = [+] = Entropy = 0

3. Green = [- , -] = Entropy = 0

4. Weighted Average = $\left(\frac{3}{6}\right)(0.918) + \left(\frac{1}{6}\right)(0) + \left(\frac{2}{6}\right)(0) = 0.459$

5. Gain(Color) = 1 - 0.459 = 0.541

ii. Shape:

1. Square = [+ , + , -] = Entropy = 0.918

2. Round = [- , + , -] = Entropy = 0.918

3. Weighted Average = $\left(\frac{3}{6}\right)(0.918) + \left(\frac{3}{6}\right)(0.918) = 0.918$

4. Gain(Shape) = 1 - 0.918 = 0.082

iii. Size:

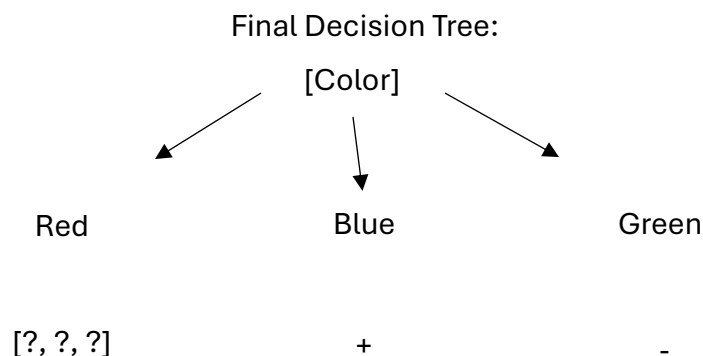
1. Big = [+ , + , + , -] = Entropy = 0.811

2. Small = [- , -] = Entropy = 0

3. Weighted Average = $\left(\frac{4}{6}\right)(0.811) + \left(\frac{2}{6}\right)(0) = 0.54$

4. Gain(Shape) = 1 - 0.54 = 0.46

d. Therefore, Color has the highest gain = split on Color first:



- i. For Red (instances 1,3,5 = +, -, +):
 - 1. Split again = try shape
 - a. Square = [+]
 - b. Round = [-, +] = still mixed = try Size
 - i. Big = [+]
 - ii. Small = [-]
- ii. Rules sample:
 - 1. If color = Blue -> class = +
 - 2. If color = Green -> class = -
 - 3. If color = Red and Shape = Square -> class = +
 - 4. If color = Red and Shape = Round and Size = Big -> class = +
 - 5. If color = Red and Shape = Round and Size = Small -> class = -

2. Impact of Adding New Attribute:

- a. For example, let's say we add a new attribute called "Pattern of Shirt" with values: "Checked", "Striped", and "Plain".
 - b. If this attribute has high information gain, it could become the new root of the decision tree. This might:
 - i. **Restructure the entire tree** if the new attribute splits data more effectively, **improve classification accuracy** especially if it resolves ambiguity in the original dataset, or **change rule priorities** as new branches may be created with more refined splits.
 - c. If a data scientist **misses this key attribute**, they could:
 - i. Deliver **misleading insights** to decision-makers, cause the company to make **costly production errors**, or lose business opportunities or cause **strategic missteps**—especially if the missed attribute is **highly predictive**.
3. Summary:
- a. In this report, we built a decision tree using the ID3 algorithm, guided by entropy and information gain, to determine the most effective attribute for classification. After analyzing the dataset, Color emerged as the most decisive factor in predicting the outcome, heavily influencing how the data was split. However, the inclusion of a new attribute such as "Pattern of Shirt" could dramatically shift this outcome by introducing a more informative split, potentially altering the tree's structure and improving classification accuracy. This reveals how a single overlooked feature can change the predictive power of a model. An analyst's ability to identify and incorporate such variables can be the difference between a mediocre and a highly effective model.

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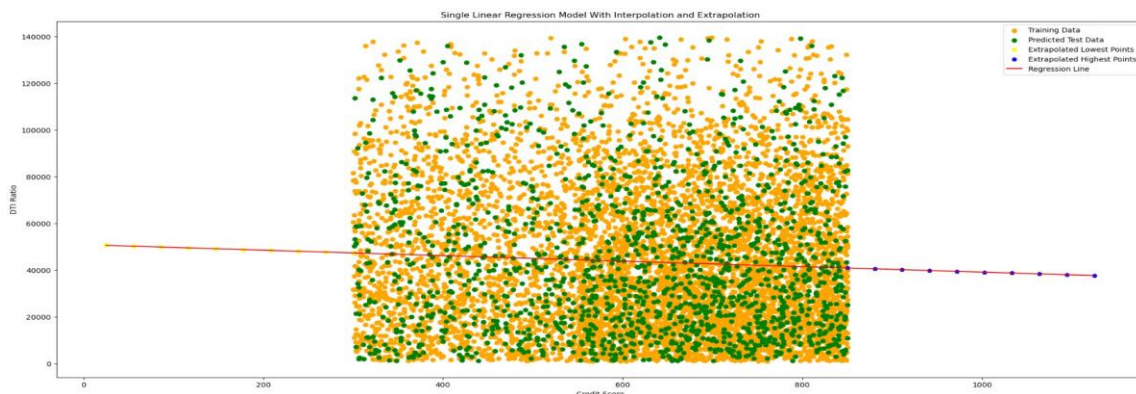
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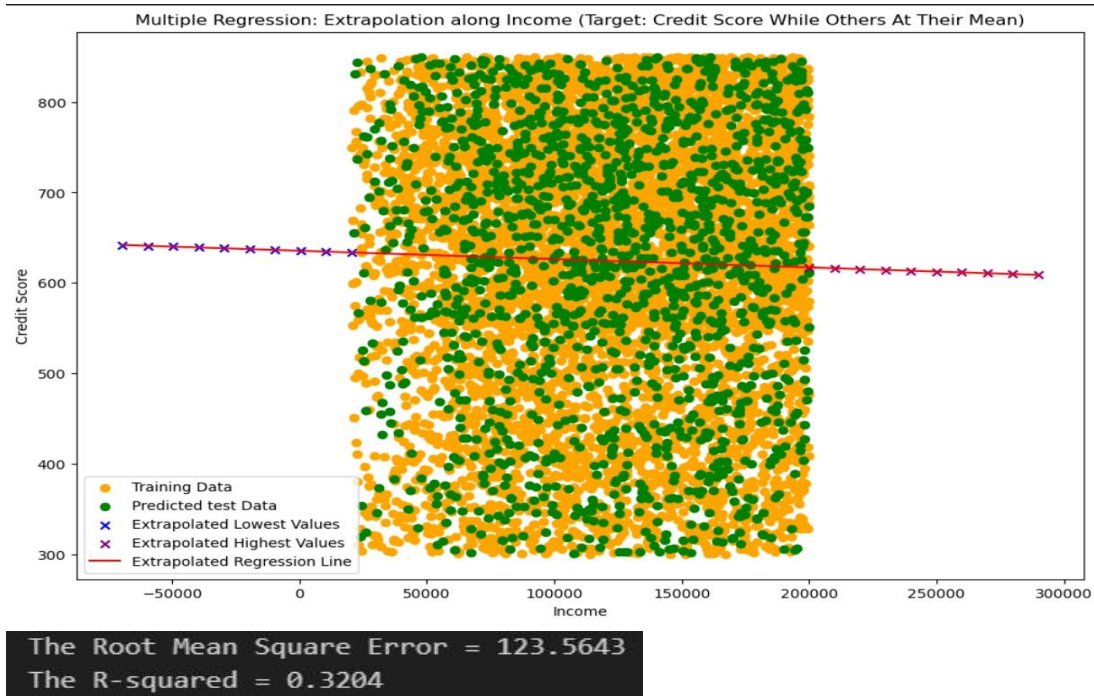
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Part B – Classification Tree Analysis (Smaller Dataset)

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Column names:

['GRE Score', 'TOEFL Score', 'University Rating', 'SOP', 'LOR', 'CGPA', 'Research', 'Chance of Admit']

Dataset Info:

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RangeIndex: 500 entries, 0 to 499

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4	LOR	500 non-null	float64
5	CGPA	500 non-null	float64
6	Research	500 non-null	int64
7	Chance of Admit	500 non-null	float64

dtypes: float64(4), int64(4)

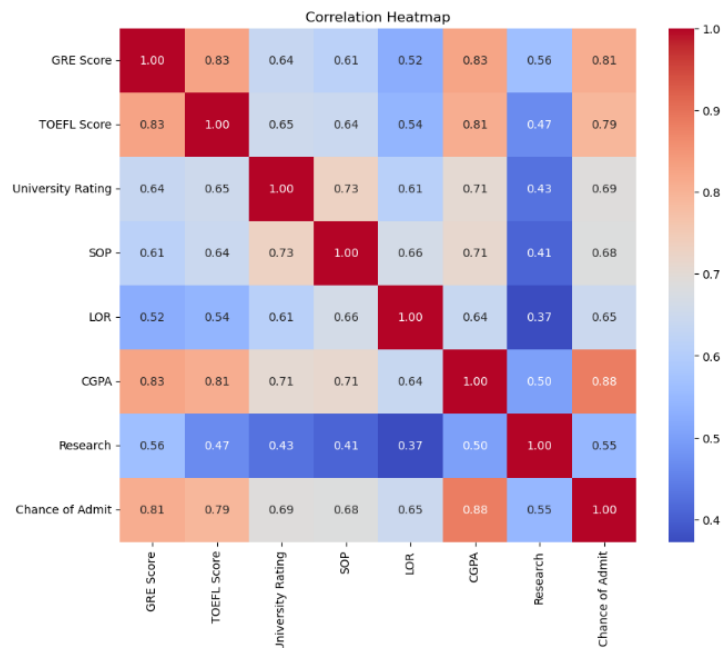
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min	6.800000	0.000000	0.34000
25%	8.127500	0.000000	0.63000
50%	8.560000	1.000000	0.72000
75%	9.040000	1.000000	0.82000
max	9.920000	1.000000	0.97000



2. Regression Analysis

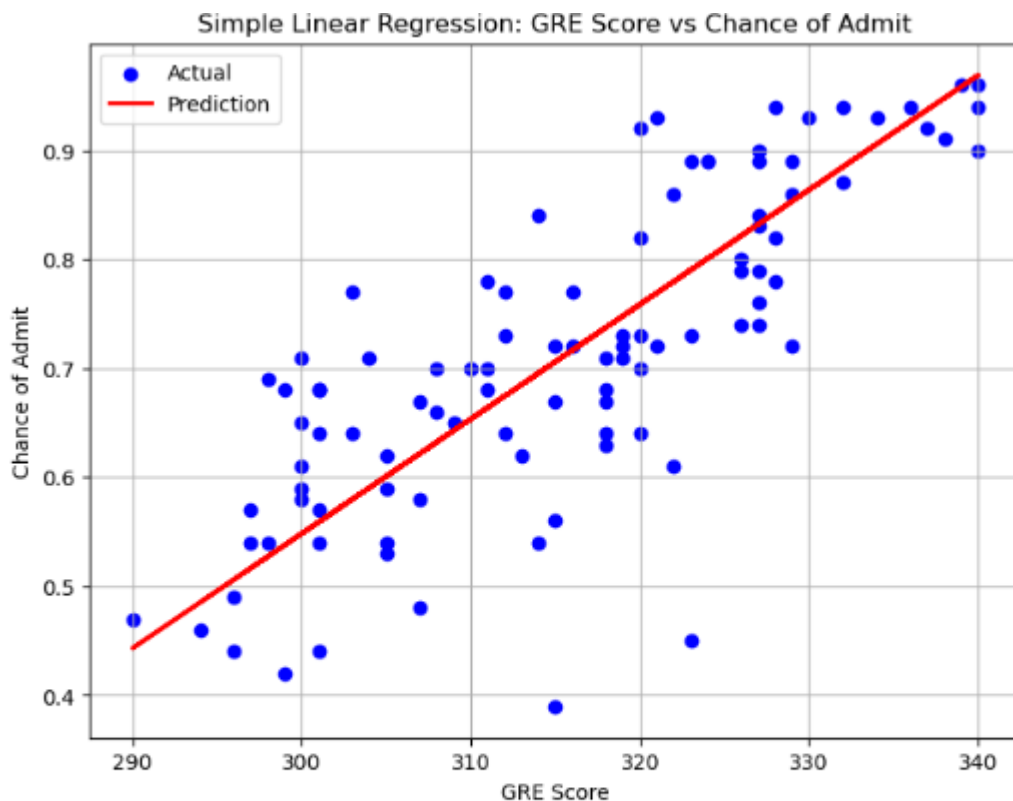
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2.2 Multiple Linear Regression

Next, a multiple linear regression model was constructed using all available features (after scaling). Although this model offered slightly improved predictions, the performance metrics confirmed that the relationship remains complex and only partially explained by the input features.



Simple Linear Regression Metrics:
MSE: 0.008464325543124892
 R^2 Score: 0.5860965504584406

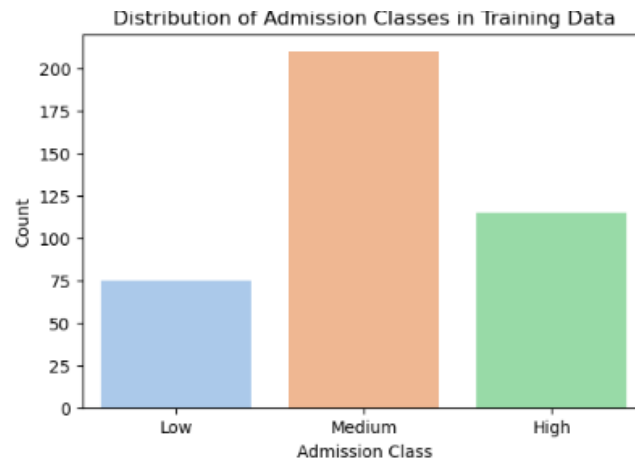
Multiple Linear Regression Metrics:
MSE: 0.0037046553987884114
 R^2 Score: 0.8188432567829628

3. Classification Tree Analysis

To offer a categorical perspective on admission outcomes, the continuous “Chance of Admit” variable was discretized into three categories:

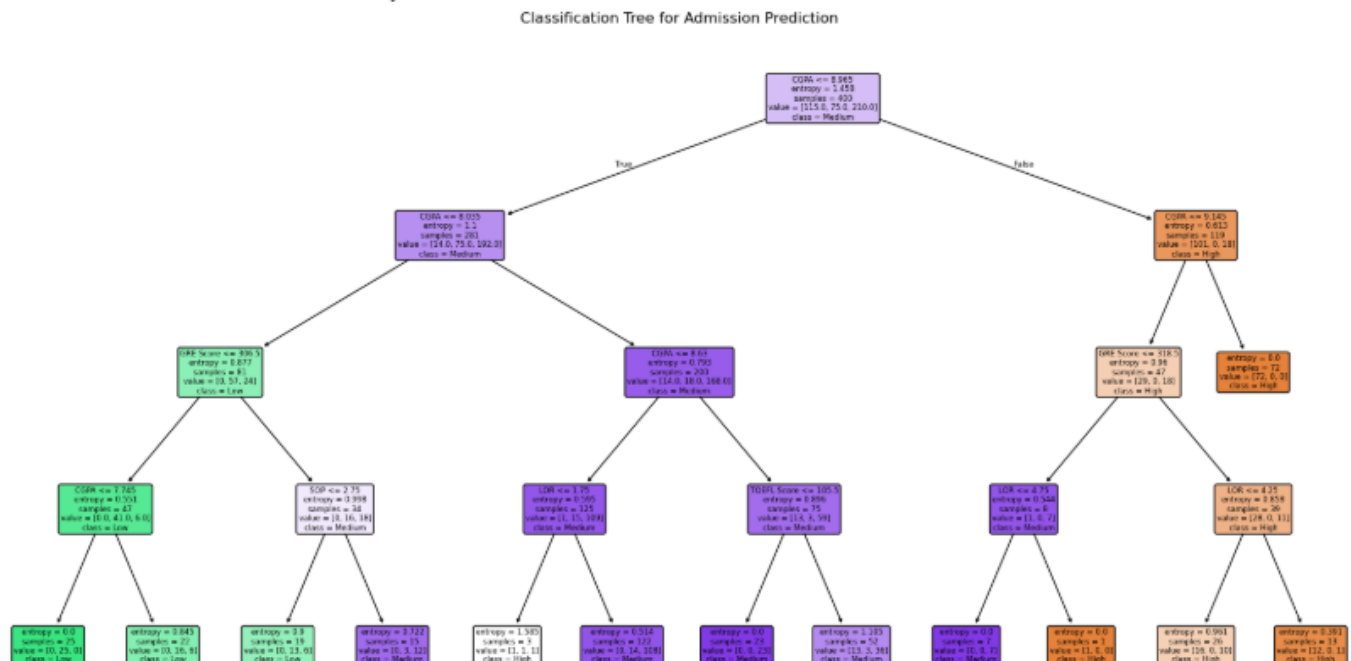
- **Low:** 0 to 0.6
- **Medium:** 0.6 to 0.8
- **High:** 0.8 to 1.0

Using these categories, we built a Decision Tree classifier. The classifier was configured with an entropy criterion and a maximum depth of 4 to maintain interpretability. The tree was trained on the original features (without scaling, as decision trees are not sensitive to feature scaling).



The resulting decision tree diagram clearly illustrates how the data is segmented. Key features such as GRE Score, TOEFL Score, and CGPA were identified as influential in classifying candidates into “Low,” “Medium,” or “High” categories.

Decision tree classifier trained successfully.



The overall performance of the classifier was evaluated using accuracy on the test set. Additionally, decision rules were extracted from the tree, offering clear insights into the model's logic.

Extracted Decision Rules from the Classification Tree:

```

|--- CGPA <= 8.97
|   |--- CGPA <= 8.03
|   |   |--- GRE Score <= 306.50
|   |   |   |--- CGPA <= 7.75
|   |   |   |   |--- class: Low
|   |   |   |--- CGPA > 7.75
|   |   |   |   |--- class: Low
|   |   |--- GRE Score > 306.50
|   |   |   |--- SOP <= 2.75
|   |   |   |   |--- class: Low
|   |   |   |--- SOP > 2.75
|   |   |   |   |--- class: Medium
|   |--- CGPA > 8.03
|   |   |--- CGPA <= 8.63
|   |   |   |--- LOR <= 1.75
|   |   |   |   |--- class: High
|   |   |   |--- LOR > 1.75
|   |   |   |   |--- class: Medium
|   |   |--- CGPA > 8.63
|   |   |   |--- TOEFL Score <= 105.50
|   |   |   |   |--- class: Medium
|   |   |   |--- TOEFL Score > 105.50
|   |   |   |   |--- class: Medium
|--- CGPA > 8.97
|   |--- CGPA <= 9.14
|   |   |--- GRE Score <= 318.50
|   |   |   |--- LOR <= 4.75
|   |   |   |   |--- class: Medium
|   |   |   |--- LOR > 4.75
|   |   |   |   |--- class: High
|   |   |--- GRE Score > 318.50
|   |   |   |--- LOR <= 4.25
|   |   |   |   |--- class: High
|   |   |   |--- LOR > 4.25
|   |   |   |   |--- class: High
|   |--- CGPA > 9.14
|   |   |--- class: High

```